Chapter 6: Jointly Distributed Random Variables (Part 1)

Sections Covered:

6.1 Joint Distribution Functions

6.2 Independent Random Variables

Why Study Jointly Distributed Random Variables?

Up to now, we mainly dealt with **one random variable at a time**.

Examples:

- Number of heads in 10 tosses
- Lifetime of a bulb
- Number of arrivals in an hour

However, many real-world situations involve **multiple random variables simultaneously**.

Examples:

Situation	Random Variables
Weather	Temperature and Humidity
Stock Market	Return of Stock A and Stock B
Manufacturing	Length and Width of a component
Queueing System	Number of arrivals and number of departures

Situation	Random Variables
Gambling	Outcome of two dice

In such cases we need to understand:

- Individual behavior of each variable
- Relationship between variables
- Probability of events involving multiple variables

This leads to the study of **joint distributions**.

6.1 Joint Distribution Functions

Definition

Suppose

- X and Y are random variables.

The **joint cumulative distribution function (joint CDF)** is defined as

$$F(x,y) = P(X \leq x,\ Y \leq y)$$

Interpretation

$F(x,y)$ gives the probability that:

- X is at most x
- AND
- Y is at most y

simultaneously.

Visual Interpretation

Think of the (X, Y)-plane.

The event

$$X \leq x, \quad Y \leq y$$

corresponds to all points lying inside the rectangle:

$$(-\infty, x] \times (-\infty, y]$$

The joint CDF is simply the probability contained in that rectangle.

Example

Let

Outcome	Probability
(1,1)	0.2
(1,2)	0.3
(2,1)	0.1
(2,2)	0.4

Find

$$F(1, 1)$$

Since

$$F(1, 1) = P(X \leq 1, Y \leq 1)$$

Only outcome:

$$(1, 1)$$

satisfies both conditions.

Thus

$$F(1, 1) = 0.2$$

Find

$$F(1, 2)$$

Need

$$X \leq 1, \quad Y \leq 2$$

Possible outcomes:

$$(1, 1), (1, 2)$$

Therefore

$$F(1, 2) = 0.2 + 0.3 = 0.5$$

Properties of Joint CDF

Every joint distribution function satisfies:

Property 1: Bounds

$$0 \leq F(x, y) \leq 1$$

because probabilities always lie between 0 and 1.

Property 2: Nondecreasing

Increasing either variable can only increase probability.

If

$$x_1 < x_2$$

then

$$F(x_1, y) \leq F(x_2, y)$$

Similarly for y.

Property 3: Limits at Infinity

Since eventually every possible value is included,

$$\lim_{x\to\infty}\lim_{y\to\infty} F(x,y) = 1$$

Property 4

If either coordinate goes to negative infinity:

$$F(-\infty, y) = 0$$

$$F(x, -\infty) = 0$$

because no values satisfy the inequality.

Obtaining Probabilities from the Joint CDF

The joint CDF contains all information about the distribution.

Any rectangle probability can be extracted.

Rectangle Formula

For

$$a < X \leq b$$

and

$$c < Y \leq d$$

the probability equals

$$P(a < X \leq b,\ c < Y \leq d)$$

$$= F(b,d) - F(a,d) - F(b,c) + F(a,c)$$

Why This Works

This is simply inclusion-exclusion.

Start with the large rectangle:

$$F(b,d)$$

Subtract:

- Left strip
- Bottom strip

Then add back the corner removed twice.

Exactly the same idea as set theory.

Joint Probability Mass Function (Discrete Case)

Suppose X and Y are discrete.

Define

$$p(x,y) = P(X = x, Y = y)$$

This is called the **joint PMF**.

Conditions

Nonnegativity

$$p(x, y) \geq 0$$

Total Probability

$$\sum_x \sum_y p(x, y) = 1$$

Example

Roll two fair dice.

Let

$$X = \text{first die}$$

$$Y = \text{second die}$$

Then

$$p(x, y) = \frac{1}{36}$$

for

$$x, y = 1, \dots, 6$$

Example Probability

$$P(X = 2, Y = 5) = \frac{1}{36}$$

Example Event

$$P(X+Y=7)$$

Possible outcomes:

$$(1,6)$$

$$(2,5)$$

$$(3,4)$$

$$(4,3)$$

$$(5,2)$$

$$(6,1)$$

Thus

$$P(X+Y=7)=\frac{6}{36}=\frac{1}{6}$$

Marginal Distributions

Often we know the joint distribution and want the distribution of one variable alone.

Marginal PMF of X

$$p_X(x)=\sum_y p(x,y)$$

Interpretation:

Add probabilities across all possible values of Y.

Marginal PMF of Y

$$p_Y(y) = \sum_x p(x, y)$$

Interpretation:

Add probabilities across all possible values of X.

Example

Joint PMF:

X	Y	Probability
1	1	0.2
1	2	0.3
2	1	0.1
2	2	0.4

Marginal of X

$$p_X(1) = 0.2 + 0.3 = 0.5$$

$$p_X(2) = 0.1 + 0.4 = 0.5$$

Marginal of Y

$$p_Y(1) = 0.2 + 0.1 = 0.3$$

$$p_Y(2) = 0.3 + 0.4 = 0.7$$

Joint Density Function (Continuous Case)

For continuous random variables we use a density.

Define

$$f(x, y)$$

such that

$$P((X, Y) \in A) = \iint_A f(x, y)\, dx\, dy$$

Conditions

Nonnegative

$$f(x, y) \geq 0$$

Total Area

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1$$

Relationship Between Joint CDF and Joint PDF

Joint CDF:

$$F(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(u, v)\, dv\, du$$

Conversely,

$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y)$$

Marginal Densities

Marginal Density of X

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$$

Marginal Density of Y

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$$

Intuition

To get distribution of X,

"sum out" all possible values of Y.

For continuous variables, summation becomes integration.

6.2 Independent Random Variables

This is one of the most important ideas in probability.

Intuition

Two variables are independent if knowing one tells you absolutely nothing about the other.

Examples of Independence

- Result of first die and second die
- Coin toss today and coin toss tomorrow
- Lifetime of one bulb and another independently manufactured bulb

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Examples of Dependence

- Height of father and son
- Temperature and electricity demand
- Stock prices in same industry

Formal Definition (Discrete)

X and Y are independent if

$$P(X = x, Y = y) = P(X = x)P(Y = y)$$

for every x, y.

Using PMFs:

$$p(x, y) = p_X(x)p_Y(y)$$

Formal Definition (Continuous)

For continuous variables,

$$f(x, y) = f_X(x)f_Y(y)$$

for all x, y.

Independence Through CDFs

Equivalent condition:

$$F(x, y) = F_X(x)F_Y(y)$$

for every x, y.

Why This Matters

Without independence:

$$P(X = x, Y = y)$$

must be computed directly.

With independence:

$$P(X = x, Y = y) = P(X = x)P(Y = y)$$

which is much easier.

Example: Independent Dice

Let

$$X = \text{first die}$$

$$Y = \text{second die}$$

Then

$$P(X = 3) = \frac{1}{6}$$

$$P(Y = 5) = \frac{1}{6}$$

Therefore

$$P(X=3, Y=5) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

Verifying Independence

To prove independence:

Step 1

Find joint distribution.

Step 2

Find marginals.

Step 3

Check whether

$$p(x,y) = p_X(x)p_Y(y)$$

for every pair.

If even one pair fails:

$$X, Y$$

are not independent.

Important Consequences of Independence

If X and Y are independent:

Product Rule for Events

For any sets A and B,

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$$

Expectation of Product

$$E[XY] = E[X]E[Y]$$

This result will become extremely important later in covariance and correlation.

Functions Remain Independent

If X and Y are independent,

then

$$g(X)$$

and

$$h(Y)$$

are also independent.

Example:

If

$$X, Y$$

are independent,

then

$$X^2$$

and

$$\sin(Y)$$

are also independent.

Summary of Part 1

Joint Distribution Function

$$F(x, y) = P(X \leq x, Y \leq y)$$

describes behavior of two random variables simultaneously.

Joint PMF

$$p(x, y) = P(X = x, Y = y)$$

for discrete variables.

Joint PDF

$$f(x, y)$$

for continuous variables.

Marginal Distributions

Discrete:

$$p_X(x) = \sum_y p(x, y)$$

Continuous:

$$f_X(x) = \int f(x, y) dy$$

Independence

Discrete:

$$p(x,y) = p_X(x)p_Y(y)$$

Continuous:

$$f(x,y) = f_X(x)f_Y(y)$$

CDF form:

$$F(x,y) = F_X(x)F_Y(y)$$

Key Idea

Joint distributions describe multiple random variables together. Marginal distributions describe each variable separately. Independence is the special case where the joint distribution factors into the product of the marginals.